Pipe Schedule Chart

Outside diameter, wall thickness, bore and mass per metre — NPS 1/8 to 24

NPS 1/8–24
23 sizes · Sch 40 / 80
1 page

NPS NOMINAL	DN MM	OUTSIDE DIAMETER IN	OUTSIDE DIAMETER MM	SCHEDULE 40 WALL IN	SCHEDULE 40 BORE IN	SCHEDULE 40 BORE MM	SCHEDULE 40 KG/M	SCHEDULE 80 WALL IN	SCHEDULE 80 BORE IN	SCHEDULE 80 BORE MM	SCHEDULE 80 KG/M
1/8	6	0.405	10.29	0.068	0.269	6.83	**0.36**	0.095	0.215	5.46	**0.47**
1/4	8	0.540	13.72	0.088	0.364	9.25	**0.63**	0.119	0.302	7.67	**0.80**
3/8	10	0.675	17.15	0.091	0.493	12.52	**0.85**	0.126	0.423	10.74	**1.10**
1/2	15	0.840	21.34	0.109	0.622	15.80	**1.27**	0.147	0.546	13.87	**1.62**
3/4	20	1.050	26.67	0.113	0.824	20.93	**1.68**	0.154	0.742	18.85	**2.20**
1	25	1.315	33.40	0.133	1.049	26.64	**2.50**	0.179	0.957	24.31	**3.24**
1-1/4	32	1.660	42.16	0.140	1.380	35.05	**3.39**	0.191	1.278	32.46	**4.46**
1-1/2	40	1.900	48.26	0.145	1.610	40.89	**4.05**	0.200	1.500	38.10	**5.41**
2	50	2.375	60.33	0.154	2.067	52.50	**5.44**	0.218	1.939	49.25	**7.48**
2-1/2	65	2.875	73.03	0.203	2.469	62.71	**8.63**	0.276	2.323	59.00	**11.41**
3	80	3.500	88.90	0.216	3.068	77.93	**11.29**	0.300	2.900	73.66	**15.27**
3-1/2	90	4.000	101.60	0.226	3.548	90.12	**13.57**	0.318	3.364	85.45	**18.63**
4	100	4.500	114.30	0.237	4.026	102.26	**16.07**	0.337	3.826	97.18	**22.32**
5	125	5.563	141.30	0.258	5.047	128.19	**21.78**	0.375	4.813	122.25	**30.95**
6	150	6.625	168.28	0.280	6.065	154.05	**28.27**	0.432	5.761	146.33	**42.57**
8	200	8.625	219.08	0.322	7.981	202.72	**42.54**	0.500	7.625	193.68	**64.64**
10	250	10.750	273.05	0.365	10.020	254.51	**60.31**	0.594	9.562	242.87	**95.98**
12	300	12.750	323.85	0.406	11.938	303.23	**79.74**	0.688	11.374	288.90	**132.04**
14	350	14.000	355.60	0.438	13.124	333.35	**94.51**	0.750	12.500	317.50	**158.11**
16	400	16.000	406.40	0.500	15.000	381.00	**123.31**	0.844	14.312	363.52	**203.52**
18	450	18.000	457.20	0.562	16.876	428.65	**155.93**	0.938	16.124	409.55	**254.64**
20	500	20.000	508.00	0.594	18.812	477.82	**183.40**	1.031	17.938	455.63	**311.16**
24	600	24.000	609.60	0.688	22.624	574.65	**255.18**	1.219	21.562	547.67	**441.84**

WEIGHT CLASSES STD AND XS

they match Sch 40 / Sch 80 up to NPS 10, then they stop

NPS	OD IN	SCH 40	STD	SCH 80	XS	SCH 40 KG/M	STD KG/M	SCH 40 HEAVIER
10	10.750	0.365	0.365	0.594	0.500	**60.31**	**60.31**	+0.0%
12	12.750	0.406	0.375	0.688	0.500	**79.74**	**73.84**	+8.0%
14	14.000	0.438	0.375	0.750	0.500	**94.51**	**81.29**	+16.3%
16	16.000	0.500	0.375	0.844	0.500	**123.31**	**93.23**	+32.3%
18	18.000	0.562	0.375	0.938	0.500	**155.93**	**105.16**	+48.3%
20	20.000	0.594	0.375	1.031	0.500	**183.40**	**117.09**	+56.6%
24	24.000	0.688	0.375	1.219	0.500	**255.18**	**140.96**	+81.0%

STD IS NOT SCH 40 ABOVE NPS 10

Same line size, same outside diameter, two different walls. At NPS 24 both pipes are 24.000 in OD, but Sch 40 carries 0.688 in of wall while STD carries 0.375 in — the STD wall from NPS 12 all the way to 24 never changes.

NPS 24 · Sch 40 → **255.18 kg/m**
NPS 24 · STD → **140.96 kg/m**
same nominal size, **+81.0%** steel

XS has the mirror problem: it pins at 0.500 in above NPS 8 while Sch 80 climbs to 1.219 in, so at NPS 24 XS is the **lighter** of the two (186.95 vs 441.84 kg/m). The name says extra strong; the number says otherwise.

WHERE THESE NUMBERS COME FROM

Outside diameter — ASME B36.10M. Below NPS 14 it is larger than the nominal size (NPS 1/2 is 0.840 in); from NPS 14 up the two coincide exactly. mm = in × 25.4, and that factor is exact by definition.

Bore — ID = OD − 2 × wall. Every bore here is that subtraction; no separate table is involved.

Mass — π t (D − t) × 7.85 with D and t in mm; multiply by 0.6720 for lb/ft. Carbon steel at 7850 kg/m³ — 304 is 1% heavier, 316 about 2%.

NPS is not a dimension — 1/8 has a 0.269 in bore, +115% against its nominal value; NPS 12 has 11.938, or -0.5%. The number stops tracking the bore as pipe grows and ends up tracking the OD.

DN — a rounded metric label, not a measurement. DN/OD runs from 58.3% at 1/8 to 98.4% at 24, so DN is nobody's actual dimension in particular.

Shaded rows — 1/2 to 12 in, the plumbing and HVAC range.